

Trevor's problem

(Dated: October 16, 2025)

We have the average fitness function

$$\bar{W}(n, \bar{y}) = \sqrt{\frac{\gamma}{2\pi}} (a_f n + \gamma c w^2 \bar{y}) \exp\left(-\frac{\gamma \bar{y}^2}{2}\right) \quad (1)$$

The average date of breeding $\bar{y}$ is an affine function of $n > 0$: $\bar{y} = \bar{a} + \bar{b}n$.

Assume that units are chosen so that $\bar{b} = 1$. We want to compute the average breeding date $\bar{y}$ as the maximum of the average fitness function. But we have the dependence on variable n that complicates the calculation.

Because I am not sure about the interpretation of variable n , I figured out two different approaches (we have to see which one is correct, since they depend on how n is interpreted):

1. n is a parameter and we want to obtain the $\bar{y}$ that yields the maximum average fitness. If this is correct, then the optimum average breeding time will be a function of n .

Then it suffices to take the derivative and solve for the critical point condition:

$$\frac{\partial}{\partial \bar{y}} \log \bar{W}(n, \bar{y}) = \frac{\partial}{\partial \bar{y}} \left(-\frac{\gamma \bar{y}^2}{2} + \log(a_f n + \gamma c w^2 \bar{y}) \right) = -\gamma \bar{y} + \frac{\gamma c w^2}{a_f n + \gamma c w^2 \bar{y}} = 0. \quad (2)$$

This yields (we choose the positive solution of the quadratic equation):

$$\bar{y}(n) = \frac{1}{2c w^2 \gamma^2} \left(-a_f n + \sqrt{(a_f n)^2 + 4(c w^2)^2 \gamma} \right). \quad (3)$$

Then, for a given n we will have an optimal $\bar{y}$. This reflect the problem of identifiability we discussed with Stefano.

2. n has to be averaged first over its distribution (the particular form of the distribution does not matter, we can do it whatever this distribution is because the dependence on n of average fitness is linear). Let $p(n)$ be the distribution of n , and average fitness over that distribution as well (I assume that $n > 0$, but this is not essential for the calculation):

$$\begin{aligned} \bar{W}(\bar{n}, \bar{y}) &= \int_0^\infty \bar{W}(n, \bar{y}) p(n) dn \\ &= \sqrt{\frac{\gamma}{2\pi}} \exp\left(-\frac{\gamma \bar{y}^2}{2}\right) \int_0^\infty (a_f n + \gamma c w^2 \bar{y}) p(n) dn \\ &= \sqrt{\frac{\gamma}{2\pi}} \exp\left(-\frac{\gamma \bar{y}^2}{2}\right) (a_f \bar{n} + \gamma c w^2 \bar{y}), \end{aligned} \quad (4)$$

for $\bar{n} = \int_0^\infty n p(n) dn$. Now, averaging the affine function $\bar{y}(n) = \bar{a} + n$ over the distribution of n yields $\bar{y} = \bar{a} + \bar{n}$, so we can solve for $\bar{n}$ and express the fitness function in terms of $\bar{y}$ and $\bar{a}$. Then the average fitness is not a function of n anymore (note that $\bar{a}$ is a parameter):

$$\bar{W}(\bar{y}) = \sqrt{\frac{\gamma}{2\pi}} (a_f \bar{n} + \gamma c w^2 \bar{y}) \exp\left(-\frac{\gamma \bar{y}^2}{2}\right) = \sqrt{\frac{\gamma}{2\pi}} (a_f (\bar{y} - \bar{a}) + \gamma c w^2 \bar{y}) \exp\left(-\frac{\gamma \bar{y}^2}{2}\right). \quad (5)$$

Thus,

$$\bar{W}(\bar{y}) = \sqrt{\frac{\gamma}{2\pi}} (-a_f \bar{a} + (a_f + \gamma c w^2) \bar{y}) \exp\left(-\frac{\gamma \bar{y}^2}{2}\right). \quad (6)$$

Then we can take the derivative:

$$\frac{\partial}{\partial \bar{y}} \log \bar{W}(\bar{y}) = -\gamma \bar{y} + \frac{a_f + \gamma c w^2}{(a_f + \gamma c w^2) \bar{y} - a_f \bar{a}} = 0 \quad (7)$$

Solving the quadratic equation gives:

$$\bar{y} = \frac{1}{2(a_f + \gamma c w^2)} \left(a_f \bar{a} \pm \sqrt{(a_f \bar{a})^2 + \frac{4(a_f + \gamma c w^2)^2}{\gamma}} \right). \quad (8)$$

If both a_f and $\bar{a}$ are positive, then the two solutions yield positive $\bar{y}$, so we need to compute the second derivative to check which one yields a maximum. The other solution would probably correspond to a local minimum.